JM_F0 (25 °C) — provisional polarized DLS analysis report

Executive Summary

Dataset. Multi-angle PCS at 25 °C: 28 angles (15–150°), three measurements per angle — depolarized VH, polarized VV (both 2D pseudo-cross), and 3D modulated cross. The full suite runs on VV.

Main result. A large, polydisperse, weakly optically anisotropic aggregate sample. The z-average R_h is **938 nm** but at/beyond the resolution limit — the angle bootstrap spans 501–2452 nm and the CORENN peak is rail-pinned at the largest resolvable size, so the size is a **lower bound**, not a measurement. The data are deep in the **Porod regime**, so $\rho = R_g/R_h$ **is not determined**. VV and VH report the same translational dynamics; rotational diffusion is below resolution; the 2.0% depolarization quantifies a small optical anisotropy.

Confidence. The z-average r_h carries replicate/angle bootstrap intervals; leave-one-angle-out cv gives predictive r^2=0.135; the corenn inversion converged but the gcv minimum is at a λ-grid boundary.

Assumption-sensitive. Viscosity, refractive index, q-from-angle calibration, the toluene/absolute-intensity reference, the assumed sphere form factor, and the CORENN regularization all enter the reported sizes; see the Assumption Register.

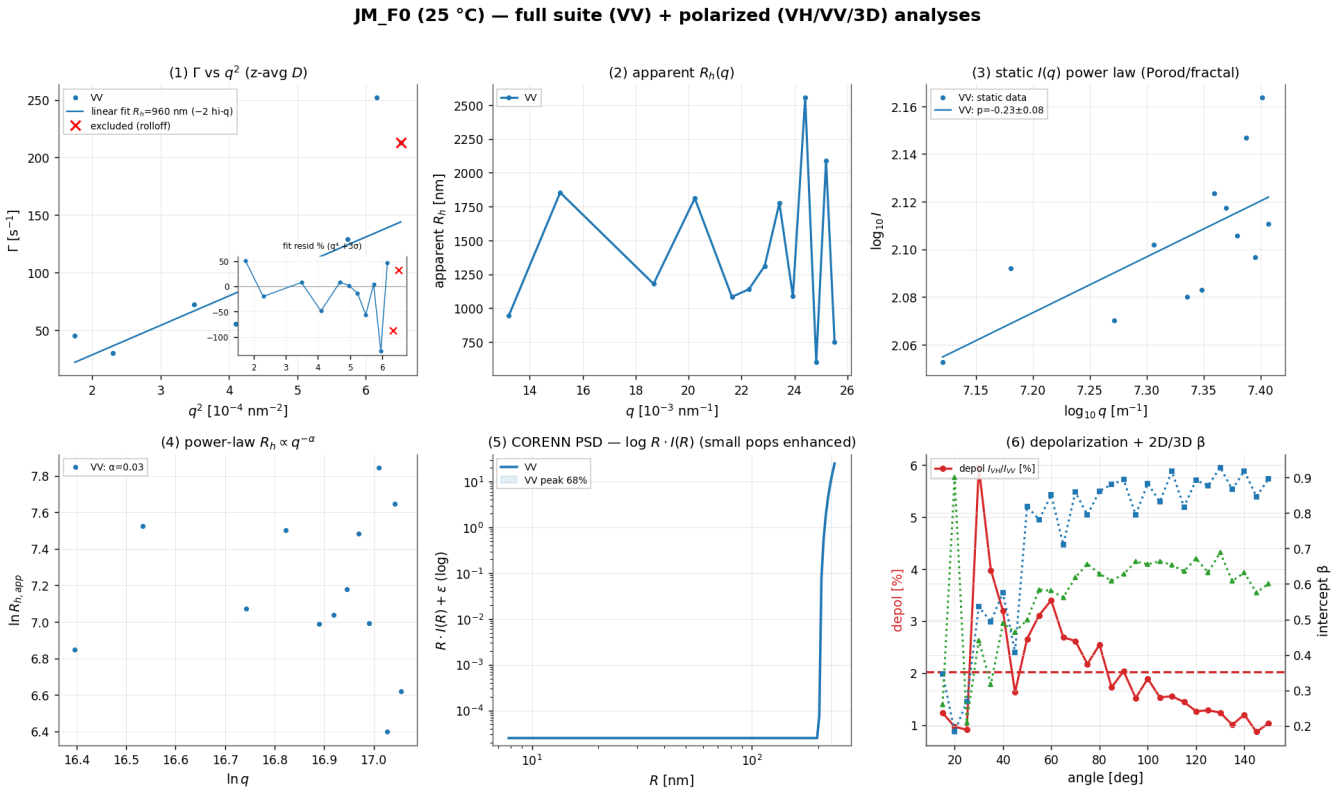
Recommended next action. measure at lower angle / use a longer wavelength to bring the μm aggregates inside the resolvable window; confirm solvent η and the toluene reference.

> **Provisional flags:** z-average size is at/beyond the q-resolution limit (lower bound); single repetition per angle — no replicate-repeatability CI (only the angle/model bootstrap); toluene/absolute-intensity reference not confirmable from the export; sphere form factor assumed (ρ not determined); ˜2% depolarization implies weak anisotropy.

Summary Table

quantity	value
angles × channels	28 × (VH, VV, 3D)
z-average R_h [68%/95% CI]	908 [68% 693–1357; 95% 501–2452] nm
shape $\rho = R_g/R_h$	not determined (Porod regime)
apparent R_h low→high q	1399→1421 nm
static power-law I∝q^{-p}	p=-0.2 (r^2=0.47)
$\Gamma(q^2)$ predictive R^2 (LOO-CV)	0.135 (in-sample 0.379)
CORENN eff.DOF / cond	12/24, κ=2e+12
CORENN multi-angle PSD	231 nm (100%)
CORENN peak [68%/95% CI, bootstrap]	231 [68% 231–231; 95% 231–231] nm
6 D_r (VH-VV intercept)	-8.7 /s (not credibly resolved)
depolarization ratio	2.02 %
mean intercept β (2D / 3D)	0.74 / 0.58

Consolidating Figures



Consolidating figure: 5 suite panels (VV) plus depolarization ratio and 2D/3D coherence β vs angle (raw-derived).

Primary Model Results

Channel identification. Per angle, the two pseudo-cross measurements are VH (low count rate) and VV (high count rate); the third is 3D modulated cross. q is recomputed from the angle. The full suite runs on the polarized VV channel.

Step 1 — per-angle cumulants → Γ(q) (VV). A 2nd-order cumulant on the initial decay (g1_min=0.6) gives the decay rate at each angle; 13 angles are kept. 15 of 28 angles are rejected because the cumulant apparent R_h exceeds the slow-mode acceptance threshold (~$3\pi/q_{\text{low}}$, here >2733 nm) — i.e. they are dominated by an unresolvably slow aggregate mode / noise.

Step 2 — Γ vs q^2 (apparent diffusion). Γ vs q^2 is linear over the bulk but the **2 highest-q angle(s) roll off** (Γ plateaus/drops at q = 0.0252, 0.0255 nm^{-1} — weak signal, number fluctuations, and residual multiple scattering at high angle), so they are excluded. A straight-line fit $\Gamma = c + D \cdot q^2$ to the remaining **11 of 13 angles** ($q \leq 0.0248$ nm^{-1}, σ from homoscedastic (residual-estimated)) gives the **apparent diffusion coefficient D = 2.555e-13 ± 3.8e-14 m^2/s** → R_h **= 960 nm** (vs 938 nm for the all-angle fit that includes the rolloff). D_{app} then varies with q for a polydisperse sample (quantified next).

Step 3 — apparent R_h(q) (polydispersity). apparent R_h is roughly flat with q, from **1399 nm** (low q) to **1421 nm** (high q) (curvature C=5.44e-16, Spearman -0.06); mean PDI 0.89 — a polydisperse population with no single dominant length scale.

Step 4 — static intensity I(q), regime from the PSD. The CORENN PSD gives dominant scatterers of R ≈ 231 nm ($R_g \approx 179$ nm, sphere-derived from the PSD radius — no Guinier regime is reached, so this is a model estimate, not a measured R_g), so q·R_g spans 2.4–4.6 across the angles → **form-factor / Porod (q·R $\gtrsim \pi$ throughout)**. On the 13 point contiguous lower-q scaling window (q=0.0132–0.0255 nm^{-1}): a near-flat / noisy high-q intensity (I ∝

$q^{+0.2}$), so no clean Porod slope (r^2=0.470) (reported as diagnostic only).

Step 5 — dynamic q-dependence. The apparent size follows a power law $R_{h,app} \propto q^{-\alpha}$ with α=0.03 (r^2=0.00) — a compact descriptor of the breadth of the distribution ($\alpha \to 0$ monodisperse, $\alpha \to 1$ fully scale-free).

Step 6 — CORENN multi-angle PSD (forward model). The joint inversion of the resolvable angles, on a radius grid bounded by dynamic observability plus the static q-window (PSD grid 8–238 nm, upper bound set by static_q) and the main-relaxation lag window (λ=1.0e+03), recovers one q-consistent intensity size distribution: peak(s) at **231 nm (100%)** — this peak sits at the PSD upper observability bound (~238 nm); the true material may extend beyond the measured lag/q window, so treat it as an unresolved slow-tail / lower-bound signature rather than a resolved peak.. An 20-angle bootstrap of the inversion puts the primary peak at 231 nm [68% 231–231; 95% 231–231 nm].

Static Regime And Shape

The CORENN PSD anticipates which static regime each angle probes (Guinier where $q \cdot R_g < 1.3$, Porod where $q \cdot R \gtrsim \pi$). When a **valid Porod slope** exists, its exponent informs the form-factor / morphology model (q^{-4} a smooth sphere; 3<p<4 a surface fractal, D_s=6-p; p≈3 a rough-surface or branched network; 2≤p<3 a mass-fractal / branched aggregate, D=p; p≈2 a coil; p≈1 a rod). $\rho = R_g/R_h$ is the model-independent shape check (sphere ≈ 0.78, coil ≈ 1.5, rod $\gtrsim$ 2), determined only when a static Guinier R_g exists.

condition	$q \cdot R_g$ span	regime	static fit	implied morphology	$\rho = R_g/R_h$
25 °C	2.4–4.6	form-factor / Porod ($q \cdot R \gtrsim \pi$ throughout)	flat/rising $I \propto q^{+0.2}$ (no clean slope)	—	not determined

Particle model. The PSD assumes a homogeneous-sphere form factor. No valid Porod slope and ρ is **not independently determined** (no Guinier regime, or a low-q structure factor), so the sphere assumption is not confirmed; treat the PSD radii as model-dependent.

Error Propagation And Sensitivity

Headline R_h and the PSD peak carry bootstrap intervals (ρ where determined). The **replicate** bootstrap is measurement repeatability; the **angle/model** bootstrap is robustness to angle selection — the dominant term for polydisperse samples.

condition	R_h replicate 68%	R_h angle/model 68%	PSD peak 68%	ρ 68%
25 °C	—	693–1357 nm	231–231 nm	—

Systematic sensitivity (not included in the intervals above). $R_h = kT/(6\pi\eta D)$ scales inversely with viscosity, so a viscosity error propagates linearly into R_h; the q calibration (scattering angle + refractive index) shifts every q-dependent quantity; the absolute-intensity (toluene) scale sets the SLS magnitude; and the assumed sphere form factor biases the PSD radii for non-spherical particles. See the Assumption Register.

Fit Quality And Model Adequacy

Beyond r^2: a hold-out cross-validation (predictive R^2), a residual-structure test (runs / lag-1 autocorrelation — structured residuals flag model inadequacy even at high r^2), and the CORENN inversion's convergence and conditioning.

condition	Γ-vs-q^2 R^2 (in / CV)	Γ resid	CORENN g2 resid	CORENN: conv / GCV / κ / eff-DOF
25 °C	0.379 / 0.135	**structured**	**structured**	yes / boundary / κ=2e+12 / 12/24

Structured Γ-vs-q^2 residuals reflect the known polydispersity curvature (the single slope is a summary, not the model); structured CORENN g2 residuals flag form-factor / single-distribution imperfection. A predictive CV R^2 well below the in-sample R^2 indicates over-fitting / poor extrapolation across angles.

Conditioning: $\kappa \approx$ 2e+12 means the inversion is severely ill-conditioned (˜12 of ˜16 double-precision digits lost), so the recovered distribution is fixed by the regularization, not the data alone — reinforcing the modality caveat. The eff-DOF/n_{basis} above is the multi-angle inversion's data-supportable complexity; it is a different inversion from the single-angle stitched one whose eff-DOF (˜5–6) appears in the size_max diagnostic.

Sample Synthesis

What the independent observables imply about the sample as a colloid — derived only from the scattering data above, with the inversion caveats carried through:

Physical picture. The scattering is dominated by **˜238 nm** objects (intensity-weighted); by number the population centres near **˜238 nm**, and the z-average R_h is 938 nm — best read as one broad, polydisperse distribution spanning these scales rather than well-separated species (see the size-distribution-by-weighting and size_max sections).

Hydrodynamic shape (Guinier–Porod → Stokes–Einstein). The static slope identifies the dominant scatterer as **a sphere (assumed)**, so the geometry-matched Stokes–Einstein relation is $\mathbf{D = kT/(6\pi\eta R)}$ — the radius R = R_h is **960 nm**. No resolved Porod slope, so the sphere form is assumed.

Primary unit. The number-weighted mode (˜238 nm) is the smallest resolved scatterer — plausibly the primary particle or smallest aggregate. A depolarized signal ($I_{\text{VH}}/I_{\text{VV}} \approx 2.0\%$) shows the scatterers are optically anisotropic (plate- or rod-like primaries), but rotational diffusion is not credibly resolved, so an independent shape/size from rotation is not available here.

Not determinable from this angular range. The lowest q still has $q \cdot R_g \approx 2.4$ (> 1.3), so the **Guinier regime is never reached**: an independent radius of gyration, an absolute molar mass / aggregation number, and $\rho = R_g/R_h$ cannot be measured here — only inferred. Reaching them needs $q \lesssim 0.00727$ nm^{-1} (smaller scattering angles, low-q SLS, or SAXS).

Caveats that bound these conclusions. *(1) Dilute limit / S(q):* the analysis assumes single-particle scattering; if the suspension is not dilute, a low-q structure factor (interparticle correlations or incipient gelation) can bias the Porod/fractal slope and the apparent sizes — a dilution series would test it. *(2) Inversion dependence:* the modes and the aggregation number inherit the ill-posedness and size_max / λ / form-factor sensitivity documented in the size-distribution sections. *(3) Stability:* this is a single time-point of a system that can aggregate or sediment; a repeat / time-resolved run (or checking the count rate for drift across the angle scan) is needed to establish the distribution is stationary.

Consistency across observables. The broad dynamic distribution (mean PDI 0.89, apparent R_h falling with q) matches the polydisperse population inferred from the static slope. No single observable here is contradicted by another — the picture is one broad, anisotropic, polydisperse population rather than a set of distinct species.

Mode-Specific Diagnostics

Rotational diffusion (depolarized VV/VH)

Polarized and depolarized translational diffusion: D_t(VV)=2.04e-13, D_t(VH)=2.13e-13 m^2/s. The VH-VV intercept difference $6D_r$ = -8.7 /s makes rotation **not credibly resolved**: the intercept is within noise. These are μm-scale aggregates, for which whole-particle rotation ($D_r \propto 1/\text{R}^3$) is far too slow to measure.

Optical anisotropy (depolarization ratio)

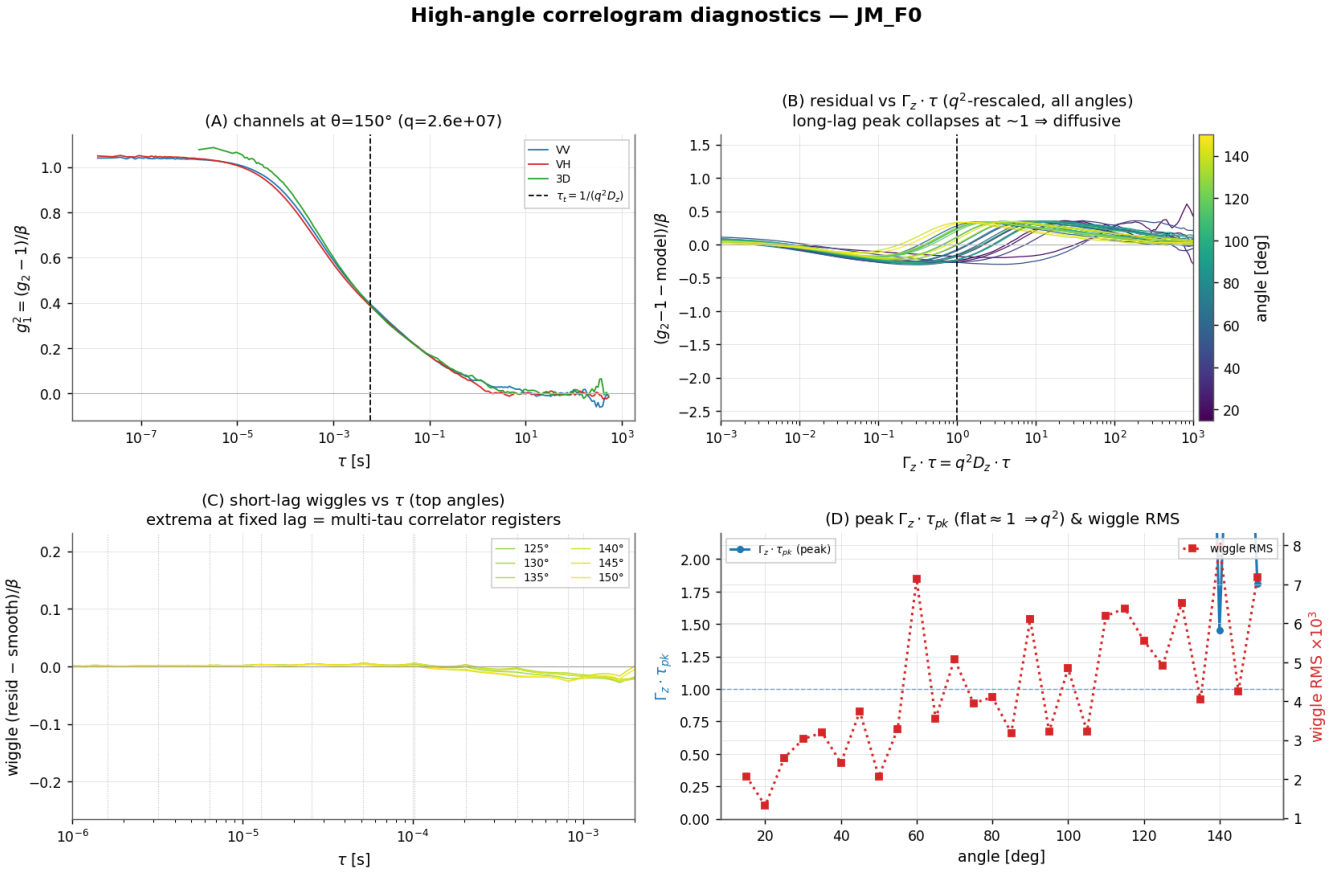
The intensity ratio $I_{\mathrm{VH}}/I_{\mathrm{VV}}$ averages **2.02 %** (0.87–5.95 % across angles) — a small but measurable optical anisotropy.

2D vs 3D cross-correlation (multiple scattering)

The 3D modulated cross-correlation suppresses multiple scattering. The coherence factor (intercept β) averages 0.74 for 2D VV vs 0.58 for 3D; their angular dependence (consolidating panel 6) flags where multiple-scattering / geometry corrections matter.

High-angle correlogram diagnostics (μs–ms ripples)

The high-angle single-exponential residual contains **two distinct features** that separate by their q-scaling: a **long-lag peak** that shifts to shorter lag with increasing angle, and **short-lag wiggles** that barely move. Panels (B)–(D) tell them apart across all angles.



Reading the panels. **(A)** VV/VH/3D at the top angle (a faster VH would be the rotational signature $\Gamma_\mathrm{VH}=\mathrm{q}^2 D_t+6D_r$; a VV-vs-3D split flags modulation / multiple scattering). **(B)** residual vs the q^2-rescaled lag $\Gamma_z{\cdot}\tau$ — a feature that **collapses onto $\Gamma_z{\cdot}\tau{\approx}1$** for every angle is a genuine diffusive relaxation. **(C)** the short-lag wiggle component vs *absolute* τ — extrema that line up at **fixed lag** (dotted: the multi-tau correlator register boundaries) are instrumental, not q-dependent dynamics. **(D)** trends: a flat $\Gamma_z{\cdot}\tau_\mathrm{pk} \approx 1$ confirms the peak scales as q^2; the wiggle RMS grows as the signal weakens at high q.

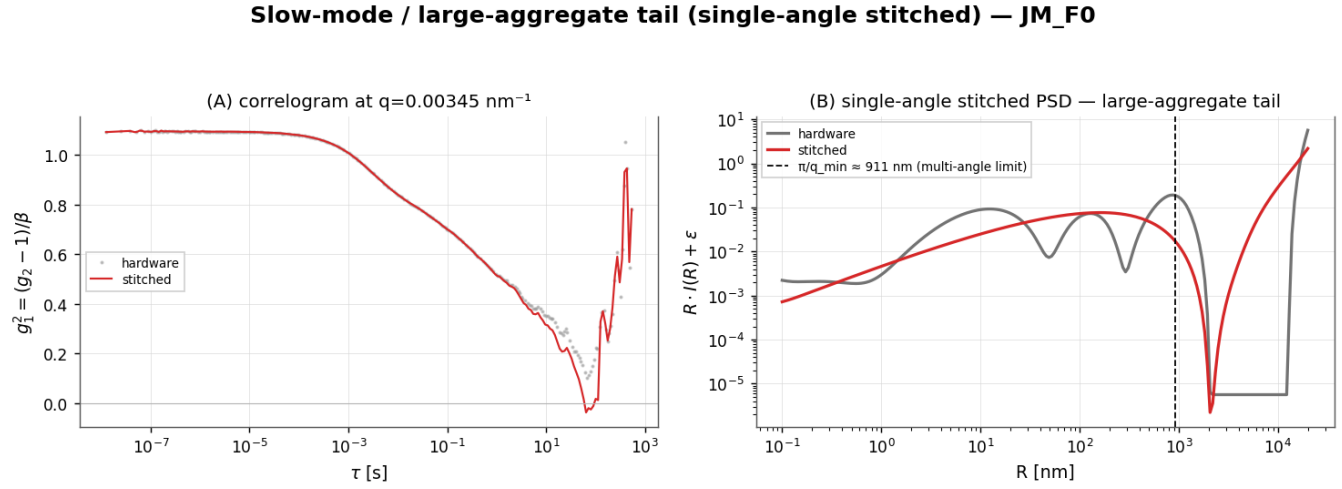
This sample — the two features separated across all 28 angles: **(1) the long-lag peak** sits at $\Gamma_z{\cdot}\tau_\mathrm{pk} \approx 3.96$ and stays roughly constant across angles (panel D), i.e. it tracks $\tau_\mathrm{pk} = 1/(\mathrm{q}^2 D_z) \propto 1/\mathrm{q}^2$ and collapses onto a single curve under q^2 rescaling (panel B) — a **genuine diffusive multi-mode / polydispersity** deviation (the broad relaxation-time distribution of the sample), strongest at high q. **(2) the short-lag wiggles** pin to *fixed* absolute lags ($\approx$ 13, 26, 51, 102 μs), independent of angle and locked to the multi-tau correlator's register boundaries (panel

C), with the amplitude growing ≈3× from low to high angle as the signal weakens — a **correlator / instrumental artifact**, not sample dynamics.

Channel check (A): at the top angle Γ_VH/Γ_VV = 1.12, i.e. VH decays faster (a rotational signature *if* not VH shot noise); but the depolarized channel is weak ($I_{VH}/I_{VV} \approx 2.0$ %) so Γ_VH is noise-limited.

Slow-mode / large-aggregate tail (single-angle stitched)

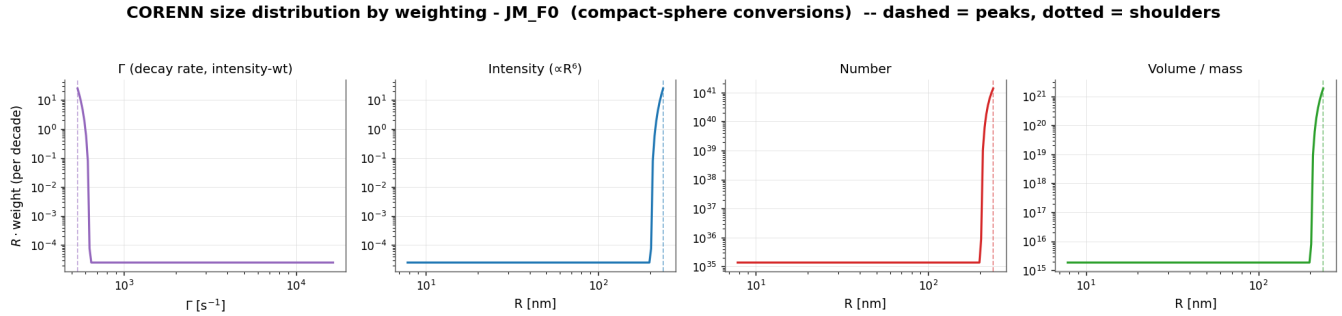
The multi-angle intensity PSD is q-resolution-limited: its R^6-weighted q→0 representation, coupled through the sphere form factor, edge-piles for R $\gtrsim \pi/q_{\min} \approx$ **911 nm**. A *single-angle* CORENN inversion of the gap-aware **stitched** correlogram (clean long-lag pooled over the full run, τ_max 537→537 s) instead recovers the slow diffusive modes directly from the decay-rate spectrum — with a data-bounded amplitude, so it resolves the large-aggregate tail where the multi-angle PSD cannot.



At the lowest angle (q=0.00345 nm^{-1}) the stitched inversion places **72% of the intensity beyond** $\pi/q_{\min}$, but it **piles at the radius-grid edge (˜20 μm)**: the aggregates here are at/beyond even single-angle resolution, so this is a **lower bound** on a large-aggregate population (a genuinely slow diffusive mode), not a measured size — consistent with this sample sitting at the q-resolution limit.

Size distribution by weighting — continuous summary, then apparent modes

DLS recovers a size distribution by an **ill-posed Laplace inversion**: many different distributions — a smooth continuum or a set of discrete modes — reproduce the measured correlation function equally well within noise, and the regularized inversion (CORENN) returns just one of them, fixed by its smoothness penalty, radius grid, and form-factor assumption. The number and sharpness of peaks are therefore properties of the inversion, not facts the data establish on their own. The report gives **two views**: (1) assumption-light continuous percentiles, and (2) the inversion's apparent modes/shoulders, flagged as inversion-dependent and trustworthy only where they survive the size_max / λ sensitivity tests. CORENN returns the *same* inversion under every weighting (the vendor's Γ / Intensity / Number / Volume tabs); re-weighting only relocates and re-weights the features.



Γ (decay-rate) view — the rawest signal (figure panel 1). Before any size model, the dominant intensity component relaxes at $\Gamma \approx$ **540 s^{-1}** (relaxation time $\tau = 1/\Gamma \approx$ **1.85 ms**) at the reference angle; the size columns below are this spectrum mapped through Stokes–Einstein $\Gamma = k_B T\ q^2/(6\pi\eta R)$.

Continuous-distribution summary (percentiles; no modes assumed). The assumption-light description — percentiles of the recovered distribution, robust to how it is partitioned into modes. Intensity is narrow (dominated by the largest scatterers); number-weighting ($\div R^6$) strongly amplifies the inverted PSD's small-R end, so its percentiles are the least certain.

Weighting	D10 (nm)	D50 (nm)	D90 (nm)	D_g (nm)	σ_g (GSD)	span	log-skew
Intensity ($\propto R^6$)	221	232	237	231	1.03	0.07	-0.91
Number (sphere, R^6)	219	231	237	230	1.03	0.07	-0.78
Volume (sphere, R^3)	220	231	237	230	1.03	0.07	-0.85

Modality caveat — read the next table as inversion fine structure. The Laplace inversion is ill-posed: a smooth continuum and a set of discrete modes can fit the same g_2 within noise, so the number and sharpness of peaks depend on the regularization (λ), the radius grid / size_max, the spline basis, and the sphere form factor — none of which the data uniquely fix. The inversion carries ≈ **12 spline degrees of freedom** — ample latitude to place peaks on a continuum. **Treat the apparent peaks/shoulders below as fine structure of this particular inversion — not established sub-populations — unless they are stable under the size_max / λ sensitivity checks and reproducible across replicates.**

Apparent modes & shoulders (CORENN fine structure — inversion-dependent)

Weighting	Comp.	Peak R (nm)	⟨R⟩ (nm)	Median (nm)	Fraction	GSD	log-skew
Intensity ($\propto R^6$)	1 (edge)	≥238	231	232	100%	1.03	-0.91 (tail → small R)
Number (sphere, R^6)	1 (edge)	≥238	230	231	100%	1.03	-0.78 (tail → small R)
Volume (sphere, R^3)	1 (edge)	≥238	230	231	100%	1.03	-0.85 (tail → small R)

An edge component piles against the radius-grid bound (≤ small-R or ≥ large-R edge) — it is unresolved, so its size is a bound only and its ⟨R⟩/median/skew are not reliable. This is typical when number/volume weighting ($\div R^6$, $\div R^3$) amplifies the inverted PSD's small-R floor.

The intensity distribution is led by a peak at **238 nm** (100% of scattered light, tail → small R); under number weighting the dominant component is at **238 nm** — the *same* data, modes relocated and re-weighted by the R-power, which is why no single mean describes it.

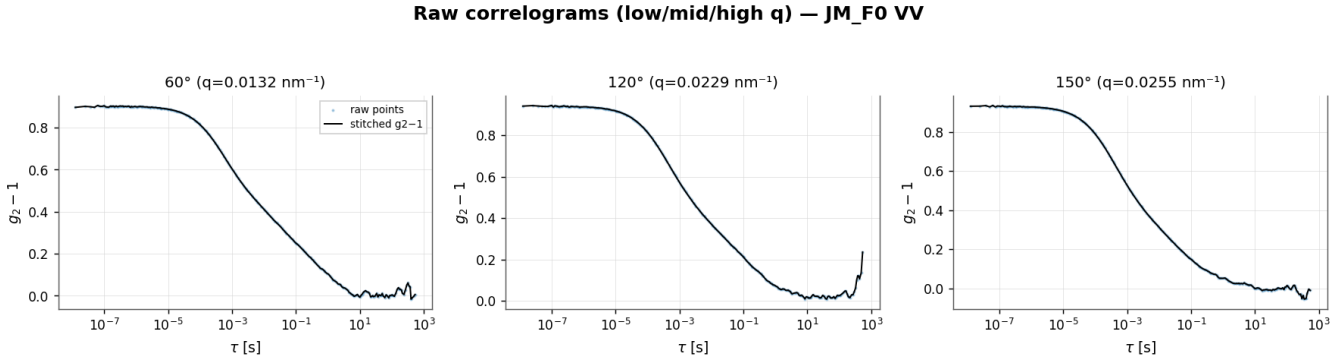
Volume/Number assume compact spheres ($V \propto R^3$); no mass-fractal correction is applied (the static slope is consistent with a compact / surface-fractal object, $D_f \approx 3$).

Data Inventory

One row per analysed condition (per-angle detail is in Per-Curve Fit Results).

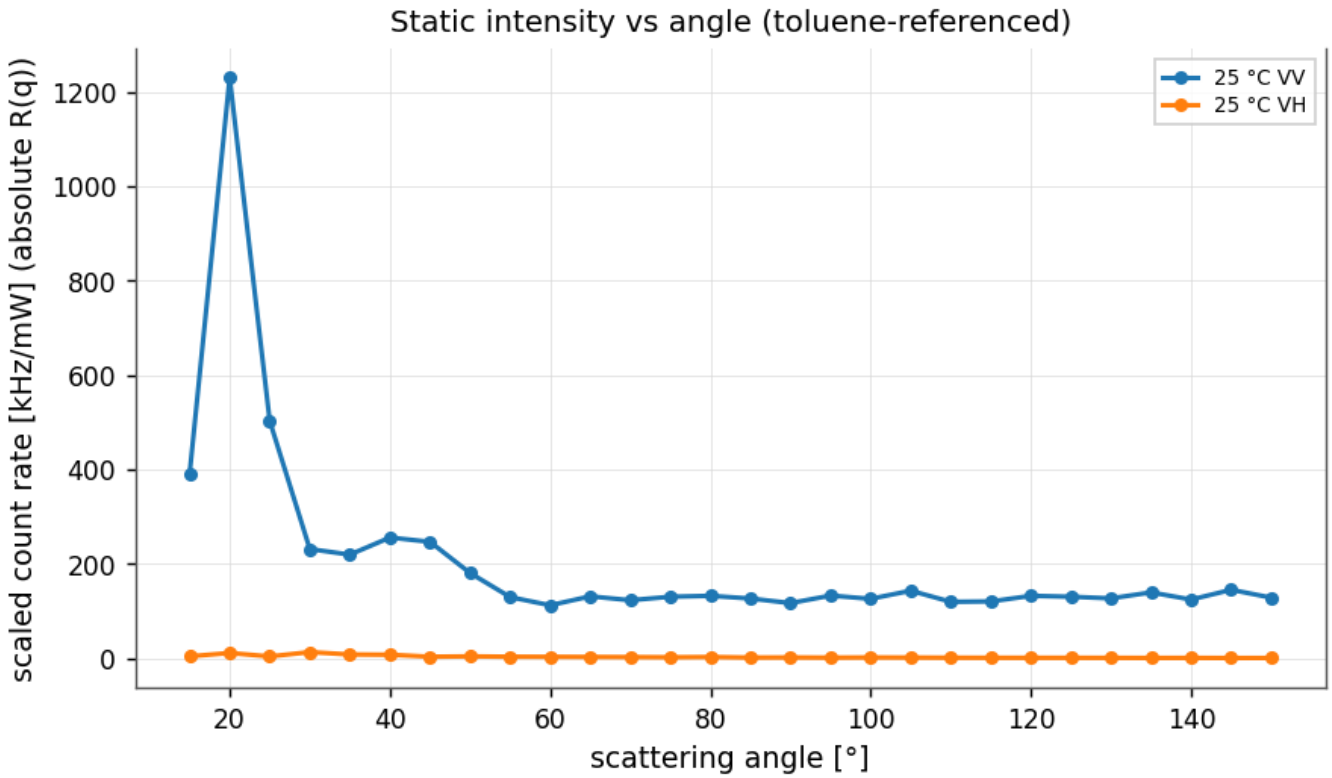
condition	channel	n angles (kept/total)	angle range	q range [nm^{-1}]	T [K]	scaled count rate [kHz/mW]	reps	missing meta
25 °C	VV	13/28	15–150°	0.0132–0.0255	298.2	113–1230	0	0

Raw Correlograms



Raw VV correlograms at low/mid/high q: the stitched multi-tau correlation (black) over the semi-transparent raw points.

Count Rates And Static Intensities (absolute, toluene-referenced)



Scaled count rate (absolute Rayleigh ratio) vs angle, VV and VH channels (raw data).

Per-Curve Fit Results

Per-curve fits — 25 °C (VV)

Per-angle 2nd-order cumulant on the leading decay (g1_min=0.6); $R_{\mathrm{h,app}} = \mathrm{kT\ q^2}/(6\pi\eta\ \Gamma)$.

angle [°]	q [nm^{-1}]	Γ [1/s]	μ_2	PDI	β	$R_{\mathrm{h,app}}$ [nm]	reps	accepted
60	0.0132	45.4	1.8e+03	0.88	0.851	943	1	yes
70	0.0152	30.4	8.1e+02	0.88	0.859	1854	1	yes
90	0.0187	72.6	4.6e+03	0.88	0.894	1179	1	yes
100	0.0202	55.5	2.7e+03	0.87	0.883	1811	1	yes
110	0.0216	106.0	1.0e+04	0.89	0.917	1083	1	yes
115	0.0223	106.9	1.0e+04	0.89	0.817	1139	1	yes
120	0.0229	97.8	8.5e+03	0.88	0.893	1312	1	yes
125	0.0234	75.8	4.9e+03	0.86	0.877	1776	1	yes
130	0.0239	129.3	1.5e+04	0.89	0.929	1088	1	yes
135	0.0244	57.2	2.8e+03	0.85	0.867	2553	1	yes
140	0.0248	251.8	6.0e+04	0.95	0.919	600	1	yes
145	0.0252	74.5	4.8e+03	0.87	0.847	2091	1	yes
150	0.0255	212.9	4.3e+04	0.94	0.896	750	1	yes

Rejected Data And Failure Diagnostics

25 °C (VV) — 15 angle(s) rejected:

angle [°]	q [nm^{-1}]	apparent R_h [nm]	reason
15	0.00345	11535	apparent R_h beyond resolvable window (>2733 nm)
20	0.00459	123393	apparent R_h beyond resolvable window (>2733 nm)
25	0.00572	13693	apparent R_h beyond resolvable window (>2733 nm)
30	0.00684	5948	apparent R_h beyond resolvable window (>2733 nm)
35	0.00794	5979	apparent R_h beyond resolvable window (>2733 nm)
40	0.00904	20765	apparent R_h beyond resolvable window (>2733 nm)
45	0.0101	3756	apparent R_h beyond resolvable window (>2733 nm)
50	0.0112	98364	apparent R_h beyond resolvable window (>2733 nm)
55	0.0122	7060	apparent R_h beyond resolvable window (>2733 nm)
65	0.0142	4949	apparent R_h beyond resolvable window (>2733 nm)
75	0.0161	3565	apparent R_h beyond resolvable window (>2733 nm)
80	0.017	2806	apparent R_h beyond resolvable window (>2733 nm)
85	0.0178	7773	apparent R_h beyond resolvable window (>2733 nm)
95	0.0195	7377	apparent R_h beyond resolvable window (>2733 nm)
105	0.021	5302	apparent R_h beyond resolvable window (>2733 nm)

Assumption Register

assumption	value	source	affects	sensitivity checked?	comment
wavelength	632.8 nm	instrument default	q, D, R_h	no	—
refractive index	1.33	assumed aqueous	q	no	replace with measured
viscosity	0.89 mPa·s	recorded export / fallback 0.89 mPa·s	R_h	yes	verify solvent and temperature
q source	recalculated from angle	pipeline	all q-dependent fits	no	vendor 'Scattering vector [1/nm]' column is mislabelled (values are 1/m); q recomputed from angle, reported in nm^{-1}
static scale	absolute (toluene-referenced) — assumed, unverified	LSII toluene reference	SLS, R_g	no	toluene reference not confirmable from parsed export
form factor	homogeneous sphere (RGD)	pipeline	PSD radii, ρ check	yes	flag if $\rho \neq 0.78$ or depolarization present
cumulant g1_min	0.6	pipeline config	Γ, PDI	yes	leading-decay window
PSD radius grid	8–238 nm	dynamic observability + static q guardrail	PSD peaks	yes	upper=min(dynamic 20108 nm, static 238 nm, default 5000 nm); limiting=static_q; tau_max=0.1 s, min field decay=0.223
PSD basis	24 B-spline control points	pipeline config	PSD smoothness	yes	—
regularization	GCV-selected λ=1.0e+03	pipeline	PSD width	yes	interior minimum required
baseline	free per fit (CORENN); leading-decay cumulant	pipeline config	Γ, PSD	yes	—

Quality Gates

Provisional report: **10 PASS, 2 WAIVE, 0 FAIL** of 12.

gate	status	note
q physically plausible / reproducible from angle	**PASS**	q recomputed from angle (vendor q column mislabeled 1/nm)
viscosity not a placeholder	**PASS**	recorded value used
fit bounds stated for every fit	**PASS**	cumulant g1_min=0.6; CORENN tau_max=0.1s, Rmax=238 nm (static_q)
rejected curves/angles listed	**PASS**	15 rejected, tabulated
residual-structure test reported (not r^2 alone)	**PASS**	runs test + lag-1 autocorrelation
headline R_h carries a bootstrap CI	**PASS**	angle/model bootstrap (no angle has $\geq$2 replicates, so no repeatability CI)
hold-out cross-validation reported	**PASS**	leave-one-angle-out
CORENN converged + well-posed regularization	**WAIVE**	converged=True, GCV interior=False, κ=2e+12
static scale confirmed absolute	**WAIVE**	toluene reference assumed (LSII default), not confirmable from parsed export
PSD basis (intensity/volume/number) stated	**PASS**	intensity-weighted primary, conversions noted
$\rho = R_g/R_h$ and particle model stated	**PASS**	not determined; sphere form factor assumed
figure captions state raw vs derived	**PASS**	captions included

Metadata

field	value
Report title	JM_F0 (25 °C) — provisional polarized DLS analysis report
Data source	data/JM_F0/25C
Report generator	reports/jm_f0_report.py (provisional)
Analyst / owner	automated pipeline
Dataset class	multi-angle, single-temperature, polarized (VH/VV/3D)
Channels present	VV
Temperatures	298.2 K
Angles	15–150° (28 per condition)
q range	0.00345–0.0255 nm^{-1}
Replicates	single curve per angle
Solvent constants	η per condition (see table); n=1.33, λ=632.8 nm (assumed aqueous)
Static scale	absolute (toluene-referenced) — assumed, unverified
Particle model	homogeneous sphere (assumed; ρ not determined; depolarization $\Rightarrow$ weak anisotropy)
Primary model	CORENN multi-angle PSD (VV) + depolarized VV/VH model
Report limitations	size at/beyond resolution limit; sphere model assumed

Reproducibility Appendix

item	value
Command	`python reports/jm_f0_report.py`
Environment	python 3.13.5, numpy 2.2.5
Git commit (working tree)	91a3b68 (dirty)
Random seeds	bootstraps seed=0 (fixed)
Generated files	JM_F0.md, JM_F0.pdf, figures/jm_f0.png, figures/jm_f0_raw.png, figures/jm_f0_slowmode.png, figures/jm_f0_weightings.png, figures/jm_f0_countrate.png, figures/jm_f0_highangle.png, JM_F0.sidecar.json